

- 4 kinds of graphs
 - directed
 - undirected
 - directed multigraphs
 - cost graphs

- vertex - vertices (vârfuluri)

X = set of vertices

- non-empty
- finite

- edges (muchii)

E = set of edges

- no duplicate edges!

- Directed graph : $E \subseteq X \times X$ (*)

the set of ordered pairs (a, b) with $a \in X, b \in X$.

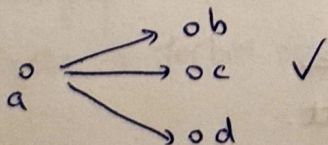
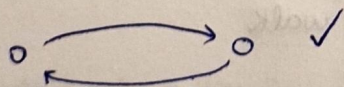
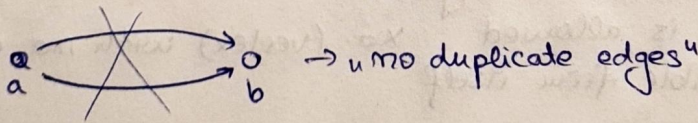
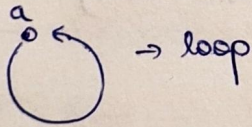
$(a, b) \in E \rightarrow$ ordered pair.

$(a, b) \neq (b, a)$

(a, a) is a loop = an edge from a vertex to itself

Loops are not formally forbidden $\Rightarrow$ definition (*) does not exclude it.

$E = \text{set} \Rightarrow$ no duplicate edges.



• Undirected graphs : $E \subseteq \{\{x, y\} \mid x, y \in X, x \neq y\}$

→ $\{x, y\} \rightarrow$ unordered pair $\Rightarrow \{x, y\} = \{y, x\}$

→ Loops are forbidden in undirected graphs ($x \neq y$)

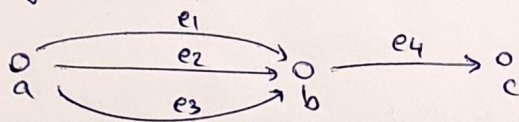
→ Useful for trees.

Tree = connected graph with no cycles.

• Directed multigraphs : $(X, E, f: E \rightarrow \underbrace{X \times X})$

→ treat edges as objects (e_1, e_2, e_3) $\hookrightarrow$ set of pairs of vertices.

→ function f tells us the endpoints ; $f(e) = (a, b)$ "edge e goes from a to b ".



$$f(e_1) = (a, b)$$

$$f(e_2) = (a, b)$$

$$f(e_3) = (a, b)$$

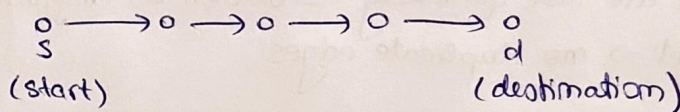
$$f(e_4) = (b, c)$$

• Walk : a sequence of vertices $(x_0, x_1, \dots, x_m)$ s.t. $(\forall) i = 0, 1, \dots, m-1$, the edge $(x_i, x_{i+1}) \in E$

$m=0$ x_0

$m=1$ x_0, x_1

↓
length



→ walks may repeat vertices and edges

→ a walk of length 0 is allowed : x_0 (vertex) with no edges $\Rightarrow$ every vertex accessible from itself.

→ length of a walk = nr. of edges along the walk
= nr. of vertices - 1.

• Path: walk with no repeating vertices or edges

→ if vertices don't repeat, edges don't repeat.

• Graph = binary relation
 → the set of edges is a relation.

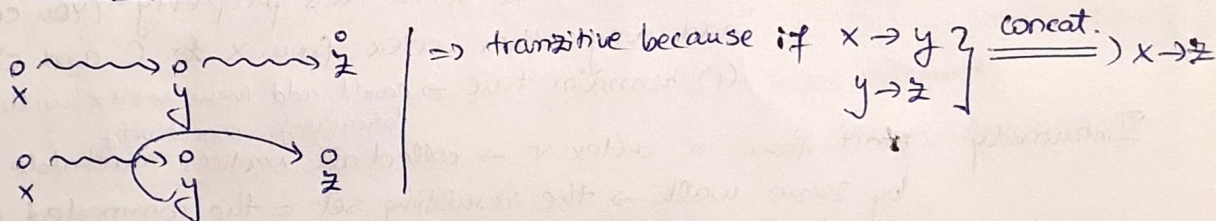
• Accessibility relation / Transitive closure of the first graph:

→ transitive because we use "walk", not "path"
 ↓
 no repeating vertices/edges.

→ "accessible": y is accessible to x if $\exists$ at least one walk starting at x and ending at y .

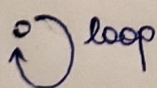
$$(x, y) \in E' \text{ iff } \exists \text{ walk } x = x_0, \dots, x_m = y$$

→ $A = \{(x, y) \mid y \text{ is accessible from } x\} \rightarrow$ accessibility relation.



• closed walk = walk having $\begin{cases} \text{start} = \text{end} \\ \text{len} \geq 1 \end{cases}$
 → can have repeating vert/edges.
 → starts and ends at the same vertex
 → length 0 $\Rightarrow$ closed walk (generally)

• Cycle = closed walk having "no repeating vertices or edges".
 → the only repeating vertex is first / last one ($x_0 = x_m$)
 → undirected graphs $\Rightarrow$ length ≥ 3 .



$$x_0, \dots, x_m$$

$$x_0 = x_m$$

$$x_0, \dots, x_{m-1} \text{ distinct.}$$

$$(x_{m-1}, x_m) \mid \rightarrow \text{this edge was repeat itself}$$

this is why we say "walk" and not "path".

! Multigraph $\rightarrow$ you can traverse an edge from u to v multiple times

! Cycle $\rightarrow$ forbid repeating edges

• Connected component $\sim$ undirected graph.

• Connected graph $\rightarrow$ each vertex is accessible from each vertex.

$\rightarrow$ "accessible" $= \exists$ walk $\Rightarrow$ for every pair u, v , $\exists$ walk $u \rightarrow v$.

• Strongly connected graph $\sim$ directed graph

$o \leftarrow o \rightarrow o$ (not strongly connected)

$(\forall) u, v \rightarrow u$ can reach v and v can reach u .

• Connected component:

$C \subseteq X$ s.t. $\rightarrow$ every vertex in C is accessible from every other vertex in C (connected subgraph) (1)

$\rightarrow C$ is maximal with this property (you cannot add any other vertex from X to C and still have (1) remain true. $\rightarrow$ can't add new vertex without breaking connectivity)

Intuitively: start from a vertex $v \rightarrow$ collect all vertices reachable from v by some walk $\rightarrow$ the resulting set = the connected comp. containing v .

• Equivalence relation (accessibility is ...)

- $\rightarrow$ reflexive ($A=A$)
- $\rightarrow$ transitive ($A=B; B=C \Rightarrow A=C$)
- $\rightarrow$ symmetric ($A=B \Rightarrow B=A$)
- $\rightarrow$ homog. rel. ($R: A \rightarrow A$) ($R \subseteq A \times A$)

$\Downarrow$
equivalence classes = connected components.

• Objects = vertices
(in this equivalence relation)

• Directed graph $\rightarrow$ strongly connected components

$a \sim b$ iff $\begin{cases} b \text{ accessible from } a \\ a \text{ accessible from } b \end{cases}$
 $\downarrow$
relation

$\Rightarrow$ this relation = equivalence relation

• Subgraphs

$\rightarrow$ map of a city is a subgraph of the map of the country.

$G=(X, E)$ a graph is determined by its vertex set and edge set.

$G'=(X', E')$; $X' \subseteq X \rightarrow$ remove vertices along with their edges.
 $E' \subseteq E$

Spanning subgraph: $X = X'$

→ Keep all vertices but maybe remove some edges

Induced subgraph: $E' = \{(x, y) \in E \mid x, y \in X'\}$

→ X' subset of vertices

→ Keep all edges from the original graph whose endpoints are both in X' .

Internal representation of a graph

→ adjacency matrix

1: $(i, j) \in E$

0: $(i, j) \notin E$

$$\begin{cases} |X| = m \\ |E| \leq m^2 \end{cases}$$

$|X| = m \Rightarrow$ adjacency matrix: $m \times m. \Rightarrow$
 $\Rightarrow$ uses space proportional to m^2 , even if
there are few edges $\Rightarrow$ problem: size.
(space comp.)

• dir. graphs: m^2 (loops allowed)
 $m(m-1)$ (loops forbidden)

• undir. graphs: $m(m-1)/2$

• Sparse graph $\Rightarrow$ lots of 0 in the adjacency matrix $\Rightarrow$ compress it.

→ Huffman idea: ~~shorter~~ frequent symbols get shorter codes.

→ $|E|$ much smaller than m^2 (adjacency lists better than adjacency matrices)

• Dense graph $\Rightarrow$ lots of 1

→ $|E|$ is close to the maximum possible (adjacency matrices are efficient)